

SHI FENG

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EDUCATION

Tsinghua University

Sep 2019 - Jul 2023 (expected)

Bachelor of Science, selected to Yao Class

- GPA: 3.87/4.0
- GRE Scores: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0
- TOEFL Best Scores: Reading 30, Listening 29, Speaking 22, Writing 29
- Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A), Research Immersion Training (A+)

RESEARCH INTERESTS

Theoretical computer science and economics, particularly algorithmic game theory, learning theory, causality, and machine learning.

RESEARCH EXPERIENCES

Multi-Agent Owner-Assisted Scoring Mechanisms

Aug 2022 - present

Advisors: Prof. [Yuhao Wang](#), Prof. [Weijie Su](#)

Tsinghua IIIS&Upenn Wharton School

- Extended owner-assisted scoring problem to a multi-agent setting and wrote a research proposal for it.
- Designed a fair, “exchangeable noise assumption free”, and truthful mechanism for the extended problem.

Peer Prediction for Learning Agents

Feb 2022 - Jun 2022

Advisors: Prof. [Yiling Chen](#), Prof. [Fang-Yi Yu](#)

Harvard EconCS Group (in-person)

- Surveyed mechanisms for information elicitation and extended the original multi-task setting to a sequential setting.
- Proved truthful convergence of correlated agreement mechanism adopted on agents using reward-based online learning algorithms and a negative result showing no-regret assumption on agents can-not guarantee truthful convergence.
- Created a peer prediction simulator and conducted numerical experiments which supported theoretical results.

Combinatorial Causal Bandits

Jul 2021 - Feb 2022

Advisor: Prof. [Wei Chen](#)

MSR Asia Theory Group

- Surveyed causal bandits algorithms for pure exploration and cumulative regret minimization.
- Presented the new combinatorial causal bandit framework with a definition of cumulative regret.
- Proposed algorithms for combinatorial causal bandits problem on binary generalized linear models with hidden variables.
- Implemented algorithms in OOP and conducted numerical experiments to show their efficiency.

Causal Inference for Social Networks

Nov 2020 - Jul 2021

Advisor: Prof. Wei Chen

MSR Asia Theory Group

- Surveyed identifiability of Bayesian causal models and completeness of the do-calculus algorithm.
- Provided transformations from an Independent Cascade Model to a Bayesian causal model.
- Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

PUBLICATIONS

(* denotes equal contribution)

Shi Feng, Wei Chen. Combinatorial Causal Bandits. **AAAI** 2023. [\[arXiv\]](#) [\[code\]](#)

Shi Feng, Fang-Yi Yu, Yiling Chen. Peer Prediction for Learning Agents. **NeurIPS** 2022. [\[arXiv\]](#) [\[code\]](#)

Shi Feng, Wei Chen. Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model. **International Conference on Computational Data and Social Networks (CSoNet)** 2021. [\[arXiv\]](#) [\[DOI\]](#)

Shi Feng*, Zimeng Song*, Weijie Su, Yuhao Wang. A Truthful Multi-Agent Owner-Assisted Scoring Mechanism. Writing in progress. [\[working paper\]](#)

SELECTED HONORS & AWARDS

Comprehensive Excellence Awards (top 10%)

Oct 2020, 2022, THU

“Star of Tomorrow” Award of Excellence

Mar 2022, MSR Asia

Best Paper Award (1/57)

Nov 2021, CSoNet

Innovation Excellence Award

Oct 2021, THU

Social Practice Excellence Award

Oct 2021, THU

“Challenge Cup” Science and Technology Competition, Top Grade Prize (top 3%)

Apr 2021, THU

Freshman Scholarship (top 10%)

Oct 2019, THU

Russian Mathematical Olympiad, Silver Medal

Apr 2019, MSE of Russia

Chinese Mathematical Olympiad, Gold Medal
National High School Mathematics League, First Prizes

Nov 2018, CMS
Sep 2016, 2017, 2018, CMS

TALKS

NeurIPS 2022, [Peer Prediction for Learning Agents](#) Oct 2022
Yao Class Seminar, [Peer Prediction for Learning Agents](#) Sept 2022
CSoNet 2021, [Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model](#) Nov 2021

SERVICES

Class Leader, Yao Class 91, Tsinghua University Aug 2022 - present
Reviewer, Journal of Combinatorial Optimization (JOCO) May 2022
Team Member, Chinese national team for Russian Mathematical Olympiad Apr 2019

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, Assembly language, Bash
Professional Applications: Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git
Languages: Chinese (native language), English (fluent)
Hobbies: soccer, road trip, handball, basketball, driving